

Covariate Status

CJ Tinant

2025-07-22 13:48:44

Contents

0.1 Overview	1
1 References	2

0.1 Overview

This document summarizes the status of covariate datasets used in the regional skew estimation project and references. All datasets are publicly available, spatially referenced, and selected for their hydrologic relevance. Data sources are listed in individual scripts in `R/01_download/`. Metadata for the datasets are provided in `docs/metadata/`. Detailed descriptions of the units, analytical resolution, scale, domain, and concept group are provided in `docs/metadata/descriptions/covariate_metadata.csv` and `data_dictionary_covariates.pdf`

0.1.1 Table 1: Folder Names and Status

Status codes track the current stage of preparation for each dataset listed above. Values range from “00 — Not Started” to “10 — Used in Modeling”. Data are stored in `data/processed/`. Status codes are described in `best-practices_reference.md`

Dataset Name	Folder Name	Status Code	Notes
EPA Ecoregions	us-ecoregions	06	NA
Koppen Geiger	koppen-climate	06	NA
MODIS 2016 NVDI	modis	06	NA
NED	ned	06	NA
NHDPlusV2.1	nhdplus_v21	06	NA
NHDPlusHD	nhdphd	06	NA
NLCD 2016 Land Cover	nlcd	06	NA
Plant Hardiness Zone	phzm	06	NA
PRISM annual pct	prism	06	NA
PRISM annual temp	prism	06	NA
PRISM daily pct	prism	06	NA
PRISM daily temp	prism	06	NA
PRISM monthly pct	prism	06	NA
STATSGO2	statsgo2	06	NA
USGS Station Data	peakflow_gages	06	NA

1 References

- Bailey, R.G., 1976a. Ecoregions of North America - Level I (Map).
- Bailey, R.G., 1976b. Ecoregions of North America - Level II (Map).
- Bailey, R.G., 1976c. Ecoregions of the United States - Level III (Map).
- Bailey, R.G., 1976d. Ecoregions of the United States - Level IV (Map).
- Beck, H.E., T.R. McVicar, N. Vergopolan, A. Berg, N.J. Lutsko, A. Dufour, Z. Zeng, X. Jiang, A.I.J.M. Van Dijk, and D.G. Miralles, 2023. High-Resolution (1 Km) Köppen-Geiger Maps for 1901–2099 Based on Constrained CMIP6 Projections. *Scientific Data* 10:724.
- Beck, H.E., N.E. Zimmermann, T.R. McVicar, N. Vergopolan, A. Berg, and E.F. Wood, 2018. Present and Future Köppen-Geiger Climate Classification Maps at 1-Km Resolution. *Scientific Data* 5:180214.
- Commission for Environmental Cooperation (Montréal, Q. and Secretariat, 1997. Ecological Regions of North America: Toward a Common Perspective. The Commission.
- Daly, C., 2013. Descriptions of PRISM Spatial Climate Datasets for the Conterminous United States. PRISM Doc 14.
- Daly, C., M. Halbleib, J.I. Smith, W.P. Gibson, M.K. Doggett, G.H. Taylor, J. Curtis, and P.P. Pasteris, 2008. Physiographically Sensitive Mapping of Climatological Temperature and Precipitation across the Conterminous United States. *International Journal of Climatology* 28:2031–2064.
- Daly, C., G.H. Taylor, and W.P. Gibson, 1997. The PRISM Approach to Mapping Precipitation and Temperature. *Proc., 10th AMS Conf. on Applied Climatology*, pp. 20–23.
- Daly, C., M.P. Widrlechner, M.D. Halbleib, J.I. Smith, and W.P. Gibson, 2012. Development of a New USDA Plant Hardiness Zone Map for the United States. *Journal of Applied Meteorology and Climatology* 51:242–264.
- Davis, W. and T. Simon, 1995. Biological Assessment and Criteria: Tools for Water Resource Planning and Decision Making. doi:10.13140/RG.2.1.4916.2726.
- Dewald, T., 2017. Making the Digital Water Flow: The Evolution of Geospatial Surface Water Frameworks. USEPA Office of Water: Washington, DC, USA.
- Dewald, T., L. McKay, L. Bondelid, J. Johnston, R. Moore, and A. Rea, 2019. User’s Guide for the National Hydrography Plus (NHDPlus) Version 2. US EPA, 182 Pp. US Environmental Protection Agency (EPA).
- Didan, K. and A.B. Munoz, MODIS Vegetation Index User’s Guide (MOD13 Series).
- Ecoregions of North America, Ecoregions of North America. <https://www.epa.gov/eco-research/ecoregions-north-america>.
- England Jr, J.F., T.A. Cohn, B.A. Faber, J.R. Stedinger, W.O. Thomas Jr, A.G. Veilleux, J.E. Kiang, and R.R. Mason Jr, 2019. Guidelines for Determining Flood Flow Frequency—Bulletin 17c. US Geological Survey.
- Gesch, D.B., G.A. Evans, M.J. Oimoen, and S. Arundel, 2018. The National Elevation Dataset. , pp. 83–110.
- Jin, S., C. Homer, L. Yang, P. Danielson, J. Dewitz, C. Li, Z. Zhu, G. Xian, and D. Howard, 2019. Overall Methodology Design for the United States National Land Cover Database 2016 Products. *Remote Sensing* 11:2971.
- McMahon, G., S.M. Gregonis, S.W. Waltman, J.M. Omernik, T.D. Thorson, J.A. Freeouf, A.H. Rorick, and J.E. Keys, 2001. Developing a Spatial Framework of Common Ecological Regions for the Conterminous United States. *Environmental Management* 28:293–316.
- Moore, R.B., L.D. McKay, A.H. Rea, T.R. Bondelid, C.V. Price, T.G. Dewald, and C.M. Johnston, 2019. User’s Guide for the National Hydrography Dataset Plus (NHDPlus) High Resolution. US Geological Survey.

- NHD Data Dictionary Quick Start Version 2.0, US Geological Survey. Omernik, J.M., 1987. Ecoregions of the Conterminous United States. *Annals of the Association of American Geographers* 77:118–125.
- Omernik, J.A., 1993. Ecoregions: A Spatial Framework for Environmental Management. Book Chapter. Environmental Protection Agency, Corvallis, OR (United States).
- Omernik, J.M., 2004. Perspectives on the Nature and Definition of Ecological Regions. *Environmental Management* 34 Suppl 1:S27-38.
- Omernik, J.M. and G.E. Griffith, 2014. Ecoregions of the Conterminous United States: Evolution of a Hierarchical Spatial Framework. *Environmental Management* 54:1249–1266.
- Primary Distinguishing Characteristics of Level III Ecoregions of the Continental United States., 2002. US Environmental Protection Agency (EPA).
- Rose, K.C., R.A. Graves, W.D. Hansen, B.J. Harvey, J. Qiu, S.A. Wood, C. Ziter, and M.G. Turner, 2017. Historical Foundations and Future Directions in Macrosystems Ecology. *Ecology Letters* 20:147–157.
- Tim Cohn, Bulletin 17B Restudy and Future Updates.
- Water Resources Council (U S.) Hydrology Committee, 1975. Guidelines for Determining Flood Flow Frequency. U.S. Water Resources Council, Hydrology Committee.
- Wiken, E.D., F.J. Nava, and G. Griffith, 2011. North American Terrestrial Ecoregions—Level III. Commission for Environmental Cooperation, Montreal, Canada 149.